

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – III

Roll No.: T038	Name: Akansha Pritam Ranade
Class: TYIT	Batch: 1
Date of Assignment: 14-08-2026	Date/Time of Submission: 14-08-2026

Part A - Theory

1. Stacked and Grouped Bar Charts

Stacked bar charts show categories divided into sub-categories within the same bar. Grouped bar charts place bars side by side for easy comparison.

2. Treemap and Pie Chart

Treemaps and pie charts are used to show part-to-whole relationships. They help compare the contribution of different categories to a total.

3. Scatter Plot

A scatter plot displays the relationship between two numerical measures. It helps identify patterns, relationships, clusters, and unusual values.

4. Dual-Axis and Combination Charts

A dual-axis chart displays two measures using separate axes. Combination charts can combine different chart types, such as a bar chart and a line chart.

5. Histogram

A histogram shows the frequency distribution of numerical data by dividing values into intervals or bins.

6. Box-and-Whisker Plot

A box plot summarizes the distribution of numerical data using the median, quartiles, and range. It can also help identify potential outliers.

7. Gantt Chart

A Gantt chart represents activities or tasks over a period of time. It is commonly used for project planning and tracking.

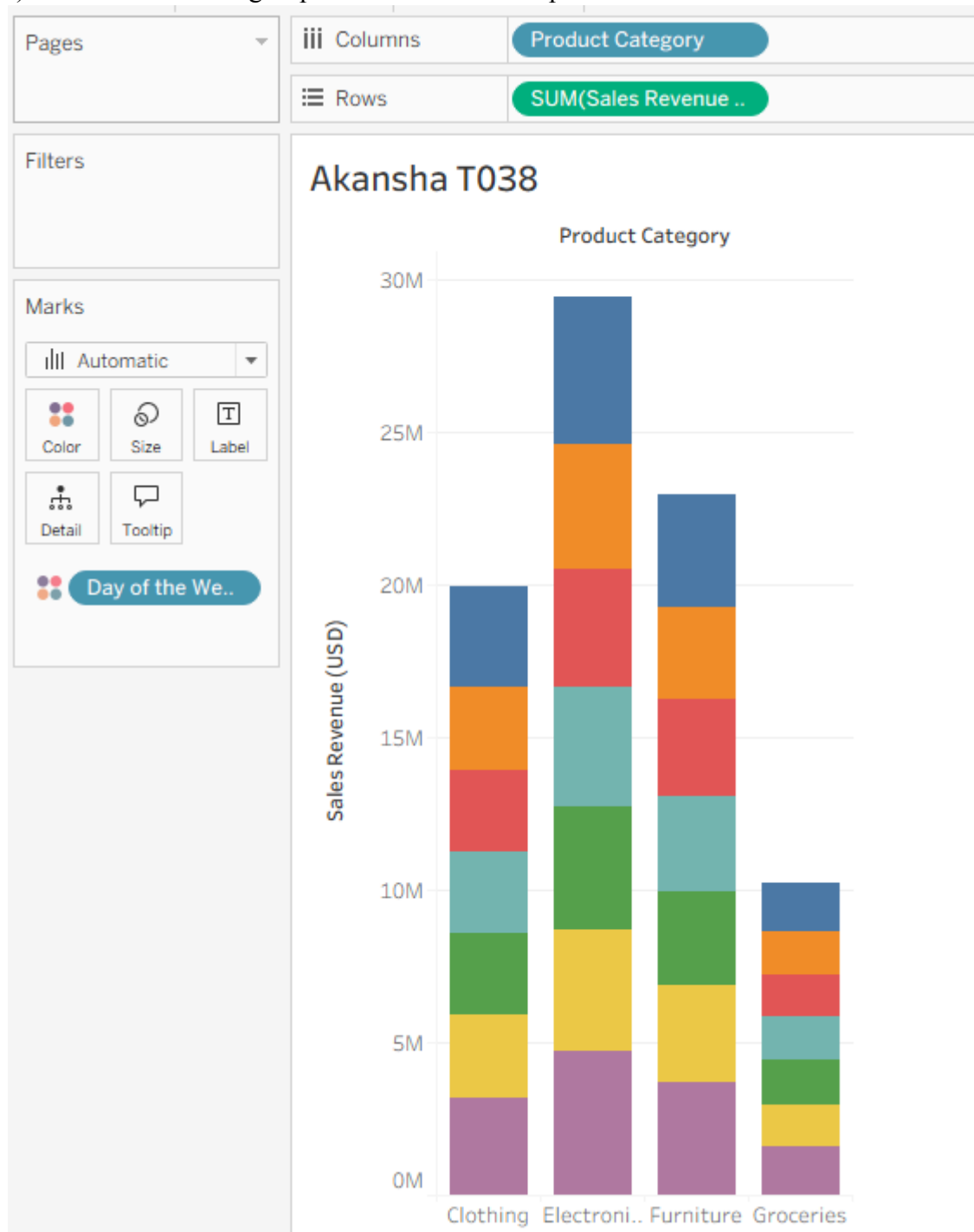
8. Advanced Date Visualizations

Tableau allows dates to be analyzed at different levels such as year, quarter, month, week, and day. This helps identify trends and changes over time.

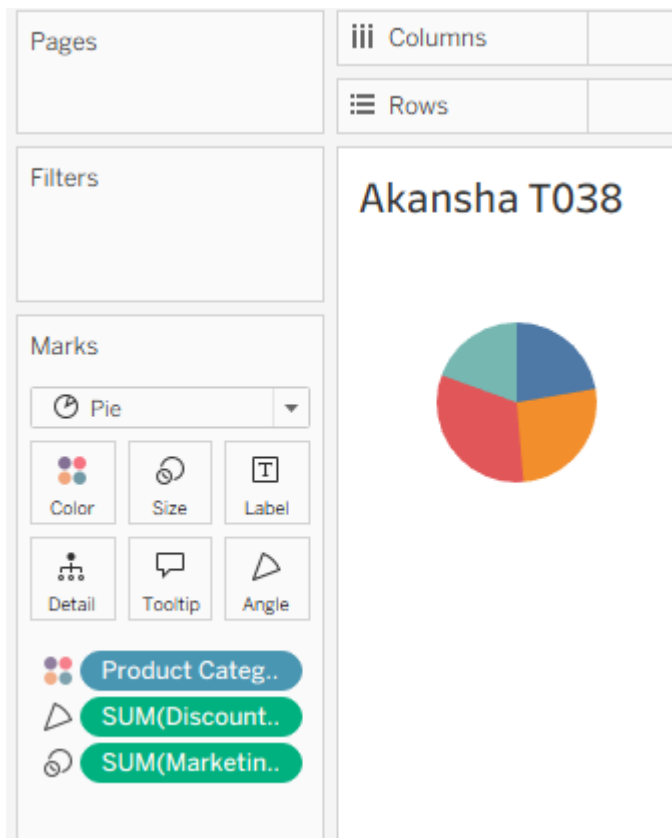
Part B – Program

Creating Advanced Visualizations in Tableau

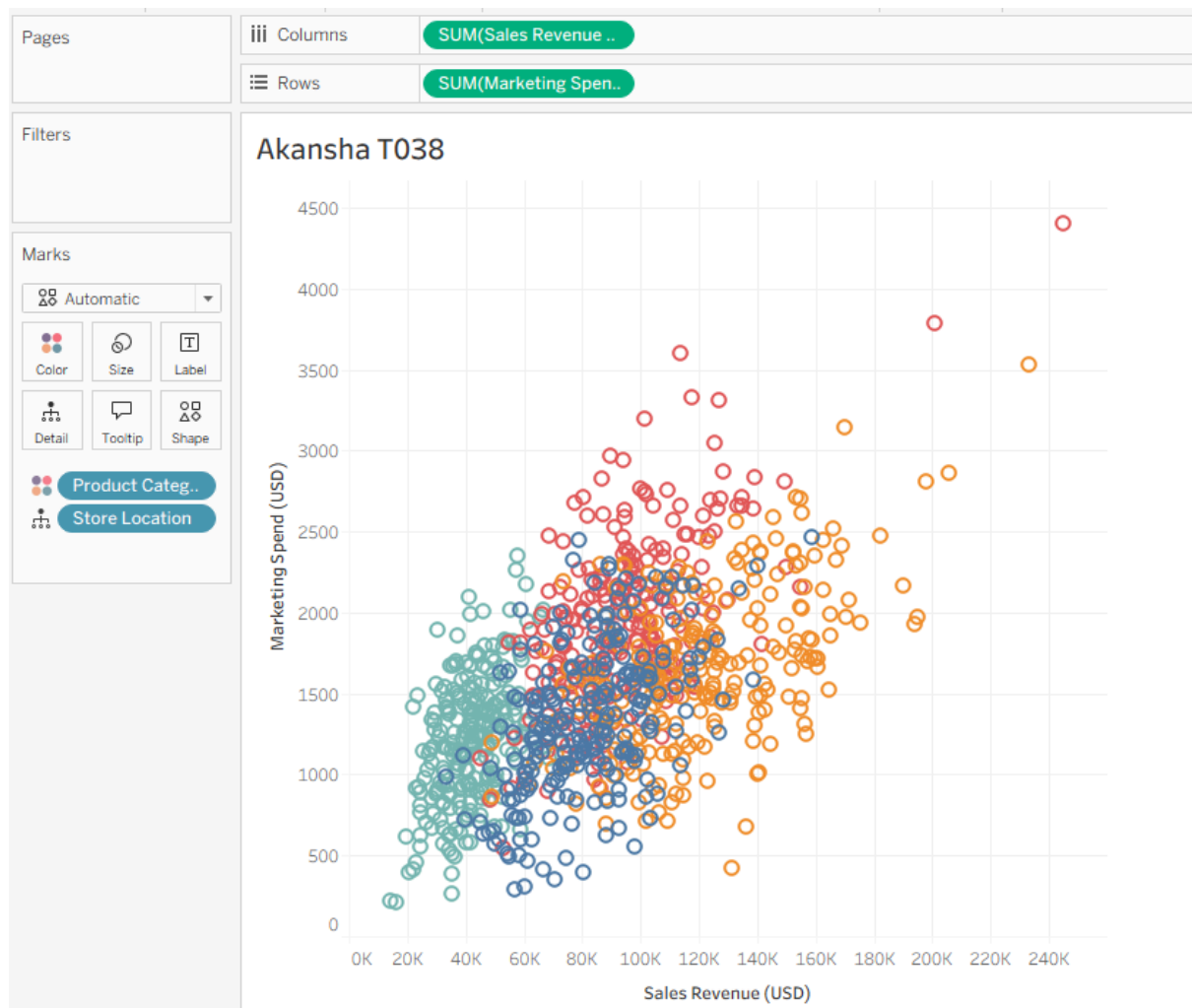
a) Create stacked and grouped bar charts for comparison



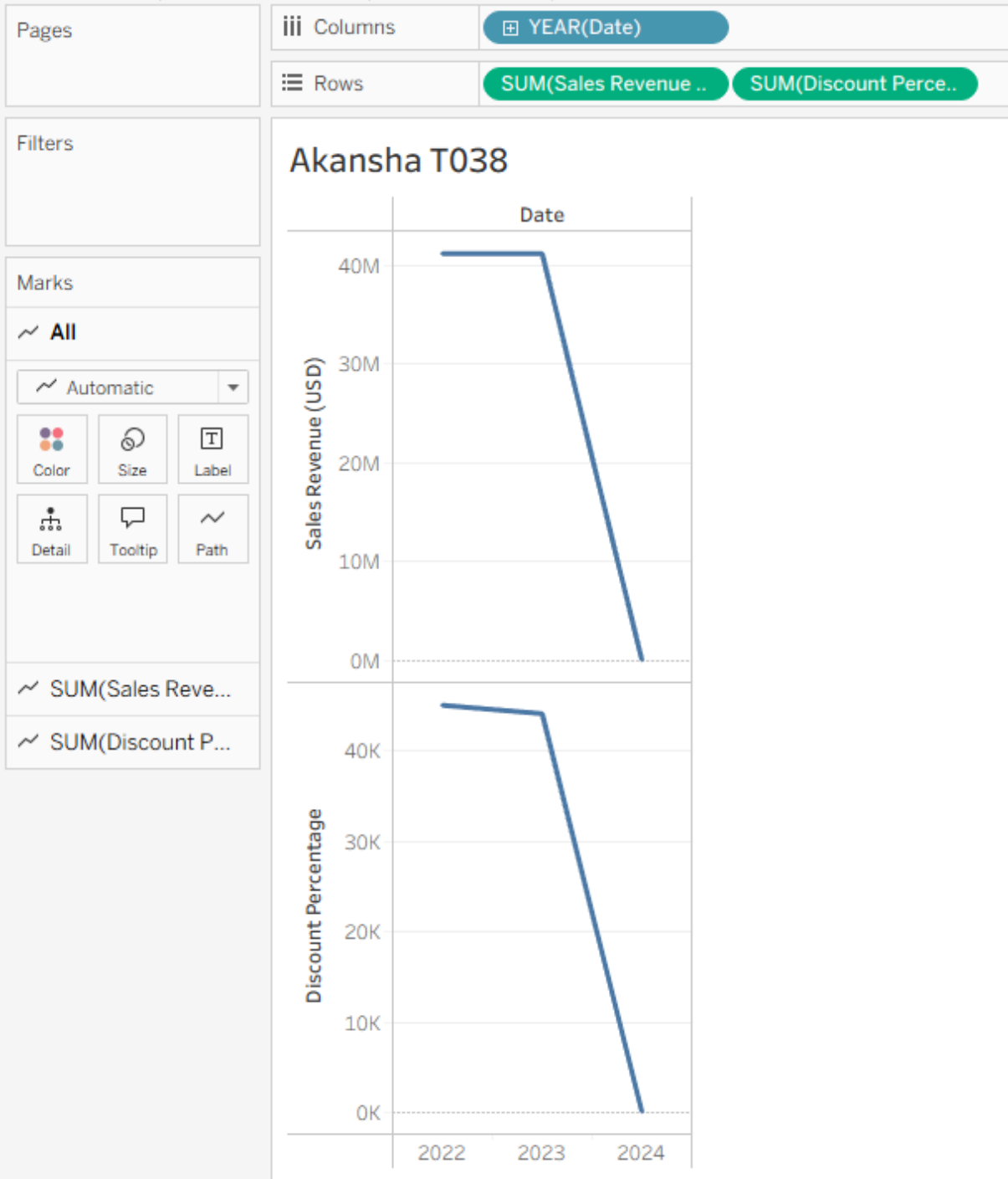
b) Build treemaps and pie charts for part-to-whole analysis



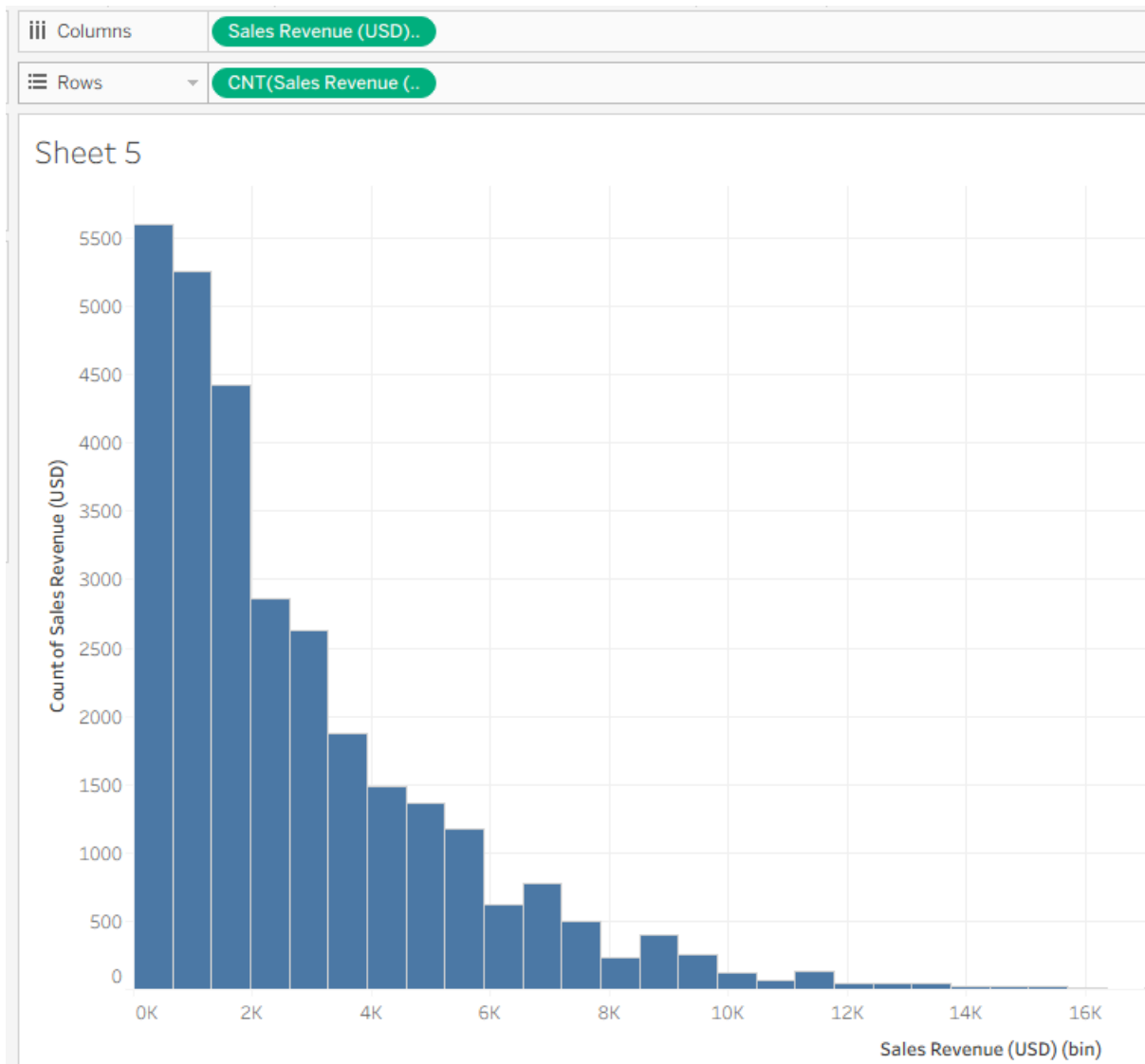
c) Design scatter plots to identify relationships



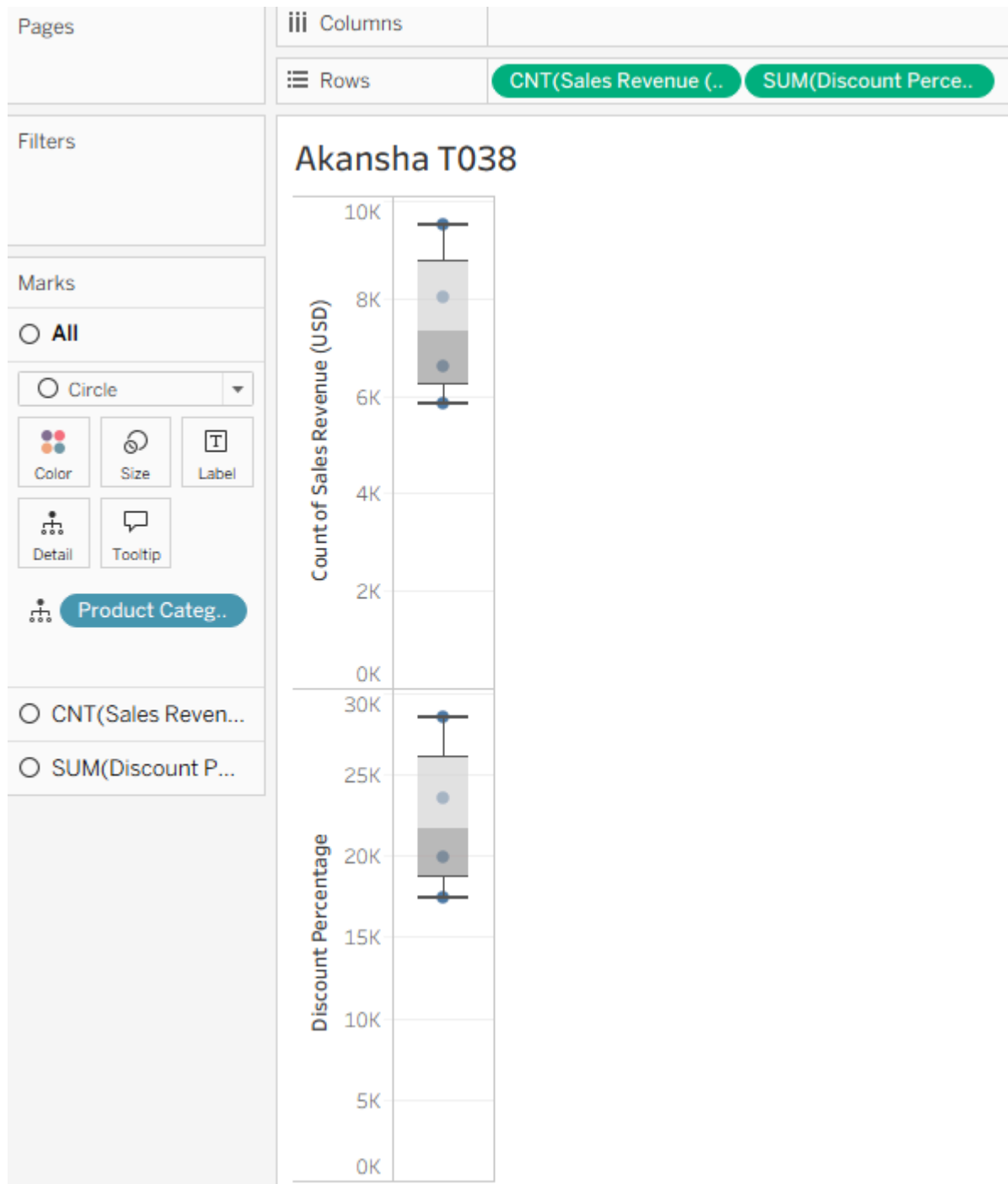
d) Create dual-axis and combination charts



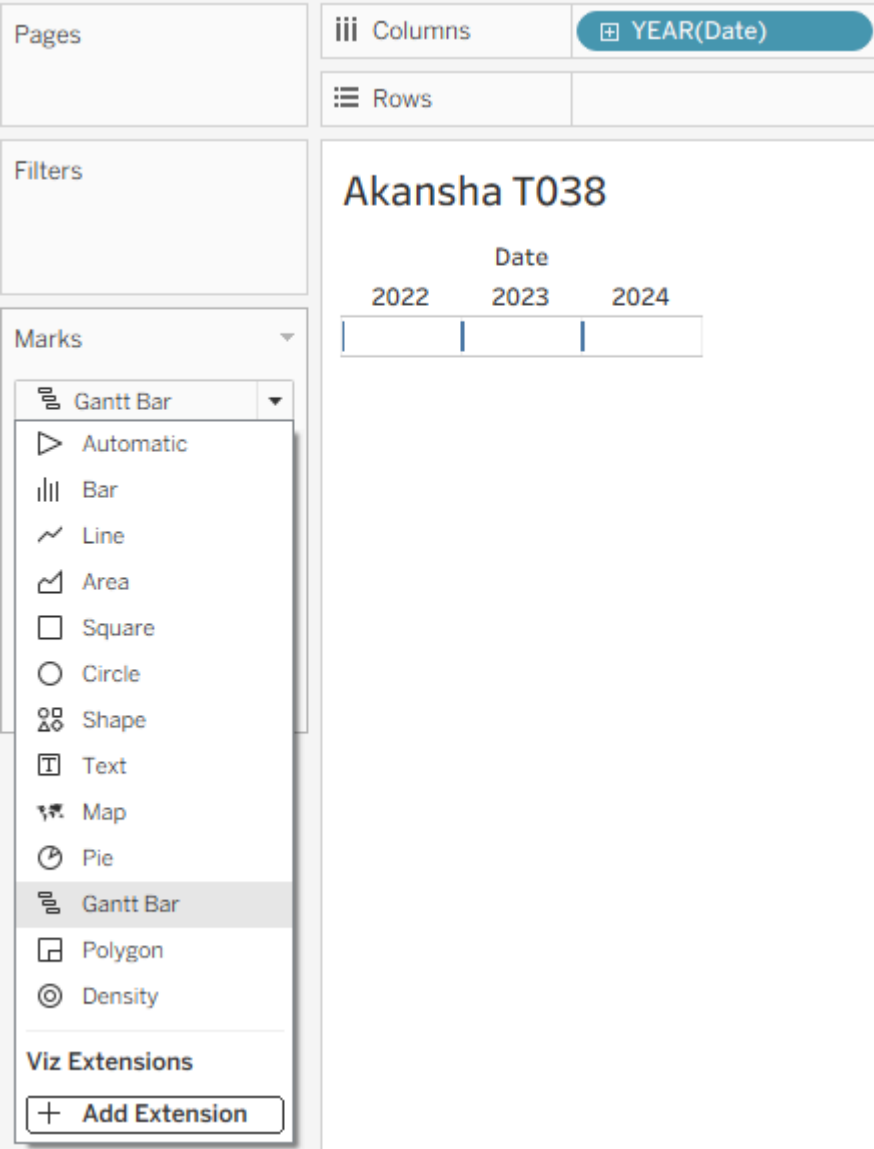
e) Develop histograms for distribution analysis

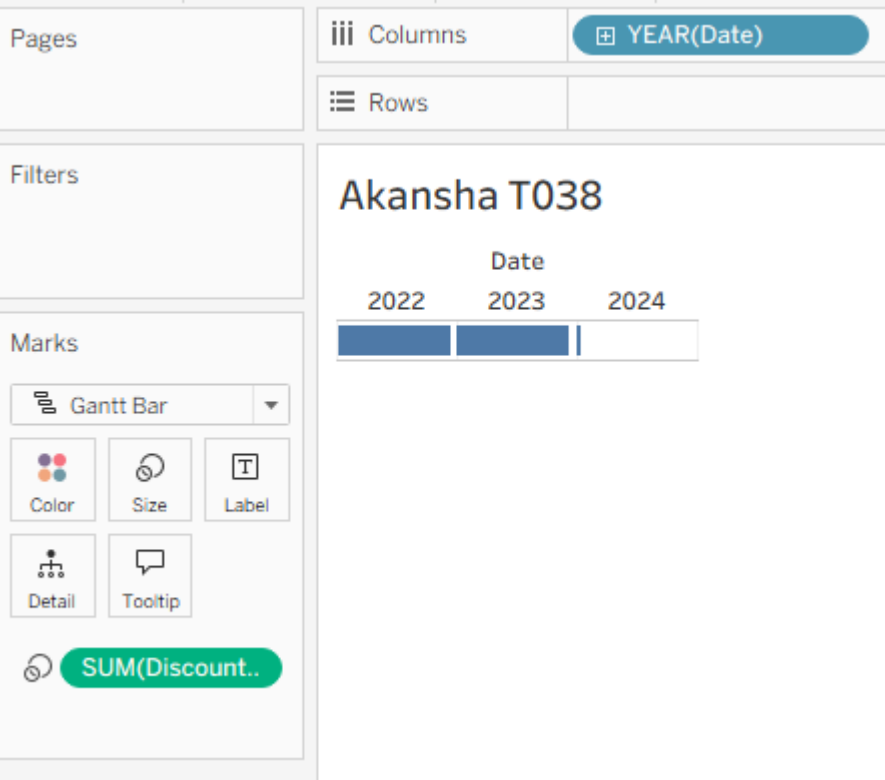


f) Build box-and-whisker plots



g) Use Gantt charts and advanced date visualizations





Part C – Curiosity Question

1. You want to compare sales of different products across regions. Would you use a Stacked Bar Chart or Grouped Bar Chart? Justify your choice.
2. A company wants to show how different product categories contribute to total sales. Which visualization would you choose and why?
3. When is a Treemap more useful than a Pie Chart for presenting part-to-whole information?
4. You want to find whether Sales and Profit have a relationship. Which visualization would you use? Explain what you would look for.
5. A manager wants to compare Sales and Profit on the same chart even though they have different scales. Which Tableau chart would you use and why?
6. A company wants to understand the distribution of customer order amounts. Which visualization would be appropriate? Explain.
7. How can a Box-and-Whisker Plot help identify unusual values in a dataset?
8. A project manager wants to show the start date, duration, and completion period of different tasks. Which visualization would you recommend and why?
9. Why would you use a Combination Chart instead of creating two separate charts?
10. A dataset contains sales records for several years. How can date fields be used to identify trends at yearly, quarterly, or monthly levels?